Learning disabilities and difficulties

A learning disability is a sign of impaired cognitive abilities and is reflected in a person's general intelligence or IQ. Learning difficulties do not affect IQ levels but make information-processing harder. Both affect how a person acquires knowledge, masters new skills, and communicates.

HOW COMMON ARE LEARNING DISABILITIES?

An estimated 1–3 percent of the world's population has some form of learning disability, and people in low-income countries are the most affected.

An intellectual or learning disability occurs when brain development is affected in some way, whether through injury or a genetic abnormality. Learning disabilities range from mild and moderate to severe and profound. The most severe may even mean that an affected person will face problems coping with independent life skills.

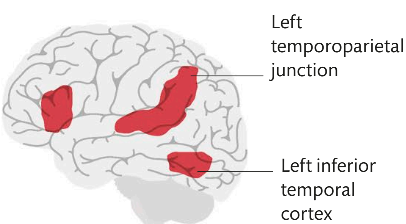
Specific causes include genetic mutations such as in Down syndrome, or fetal head injuries, maternal illness, a lack of adequate oxygen to the brain before or during birth, or brain damage from a childhood illness or injury. Some conditions have no identifiable cause. No two learning disabilities are the same, and they can include a wide variety of symptoms.

Some people with learning disabilities can talk easily and care for themselves but may take longer than usual to learn new things. Others may not be able to communicate at all. Some may also face mobility problems, heart defects, or epilepsy, which can shorten life expectancy.

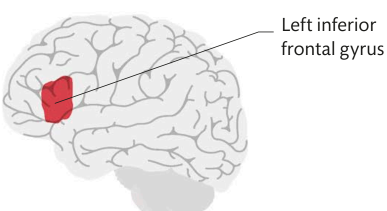
Affected people may also have associated learning difficulties—for example, someone with cerebral palsy (see p.204) may have impaired cognitive function and dyspraxia, or a person on the autistic spectrum may have a severe form of developmental delay.

Learning difficulties

Distinguishing some learning disabilities from learning difficulties can be challenging. Generally, however, learning difficulties do not affect intellectual ability or aptitude but instead impact on how the brain processes data. Someone with dyslexia, for example, which makes reading, writing, and spelling difficult, often has dyspraxia, which affects fine motor skills and coordination.



NORMAL READERS



DYSLEXIC READERS

The dyslexic brain
Areas of the brain activated during reading differ hugely in normal readers and dyslexics. Only the left inferior frontal gyrus activates in dyslexics, but this is paired with increased activity in the right hemisphere—which is why many dyslexics are highly creative.

SOME COMMON LEARNING DISABILITIES AND DIFFICULTIES

NAME	DESCRIPTION
Dyslexia	Impaired ability to learn to read and/or write. In addition to poor reading and spelling skills, dyslexics may have problems with sequences, such as date order, or difficulties organizing their thoughts.
Dyscalculia	Difficulty processing numbers, learning arithmetical concepts such as counting, and performing mathematical calculations. Dyscalculia often occurs alongside dyslexia or other learning difficulties.
Amusia	Literally meaning "lack of music," amusia is sometimes known as tone deafness and means that a person with normal hearing is unable to recognize musical tones or rhythms or reproduce them.
Dyspraxia (developmental coordination disorder)	The inability to make skilled movements with accuracy, dyspraxia is often first noticed in childhood as "clumsiness." It can cause problems with establishing spatial relationships, such as positioning objects.
Specific language impairment	Indicated by a delay in acquiring language skills where no developmental delay or hearing loss is present, specific language impairment has a strong genetic link and often runs in families.